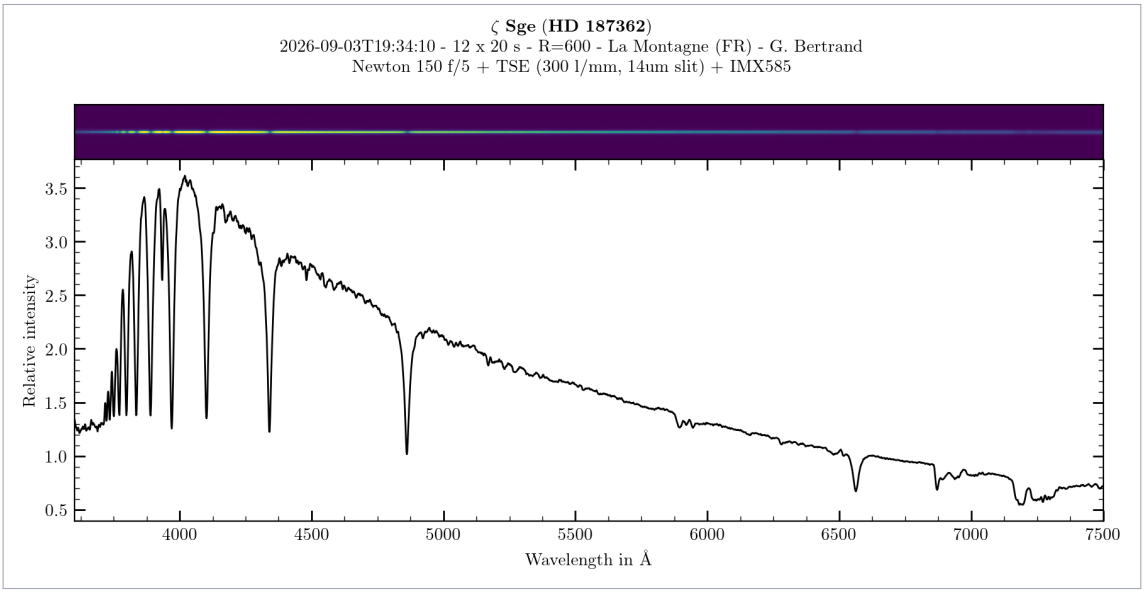
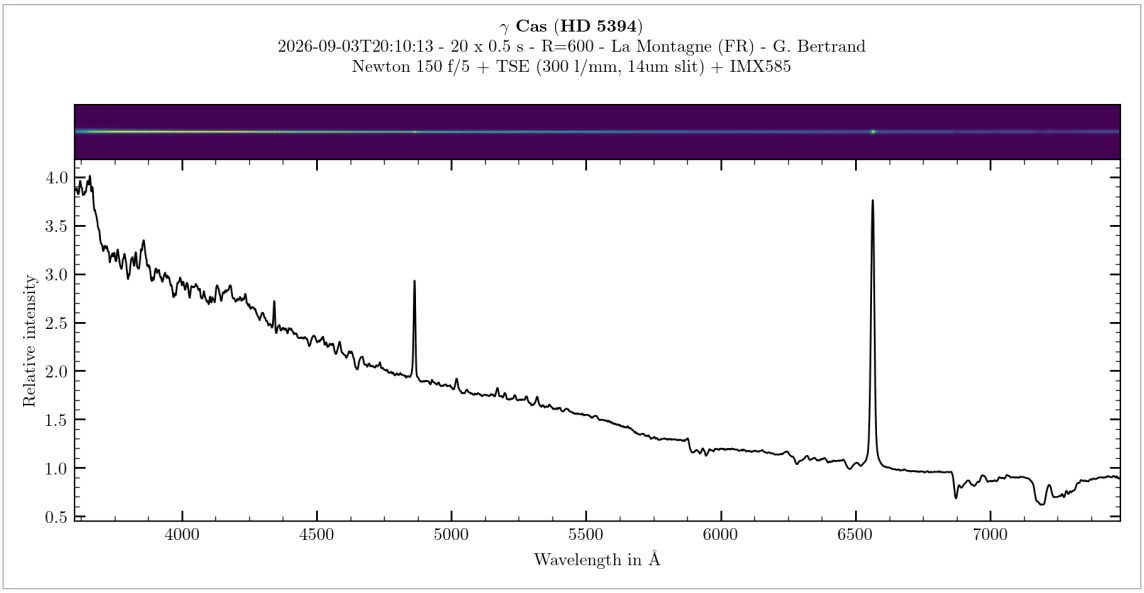
TinyStar'Ex

Examples of spectra

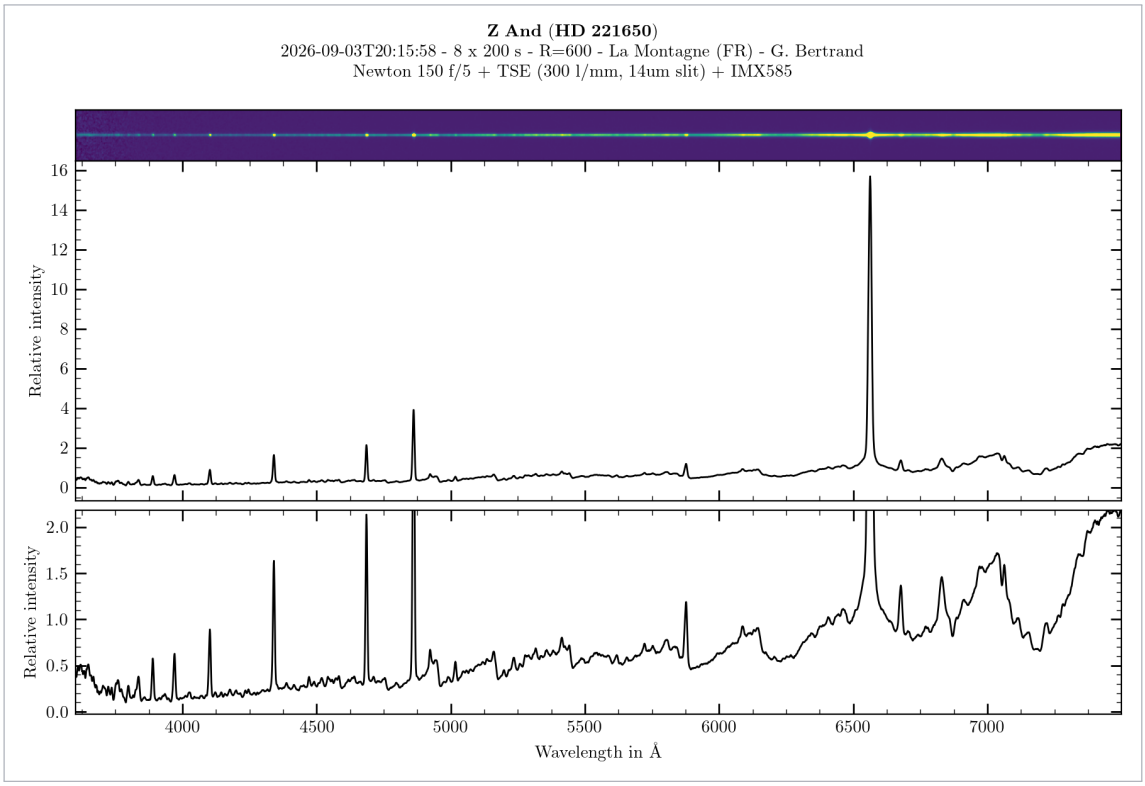
Spectrum of the star ζ Sge. TSE on a 150 mm f/5 Newtonian telescope. Guillaume Bertrand.



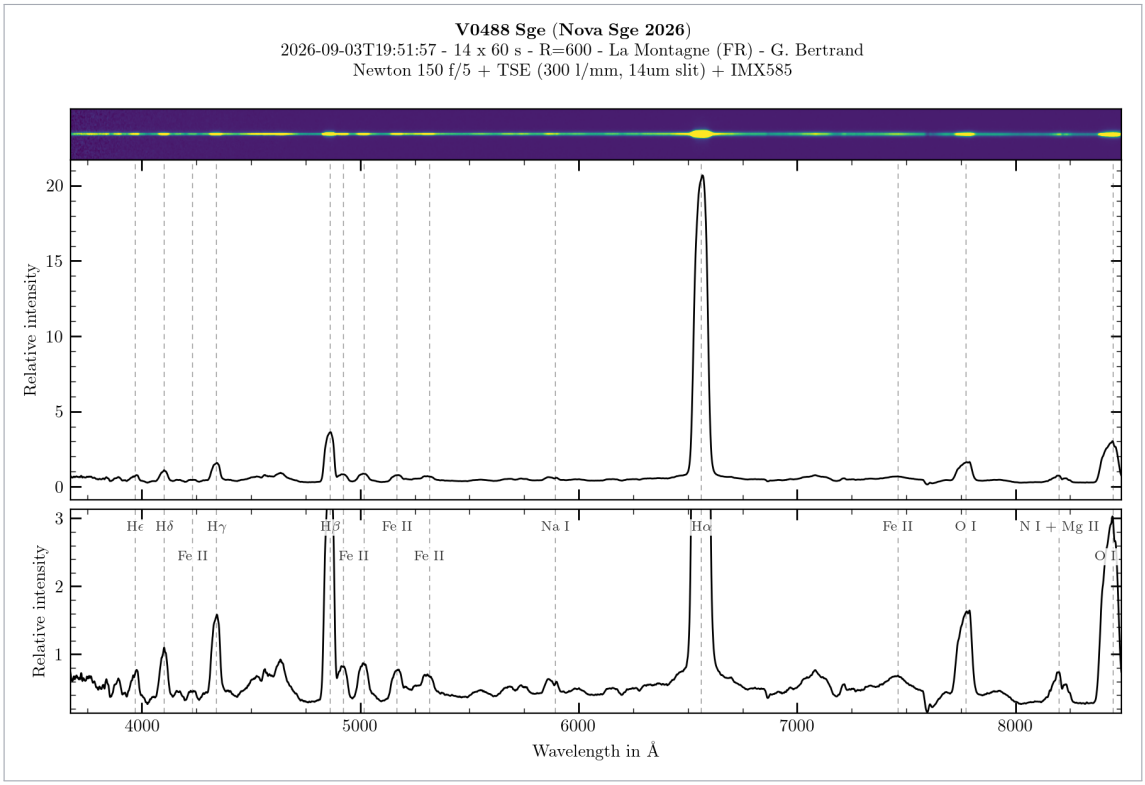
Spectrum of the Be star γ Cas. TSE on a 150 mm f/5 Newtonian telescope. Guillaume Bertrand.



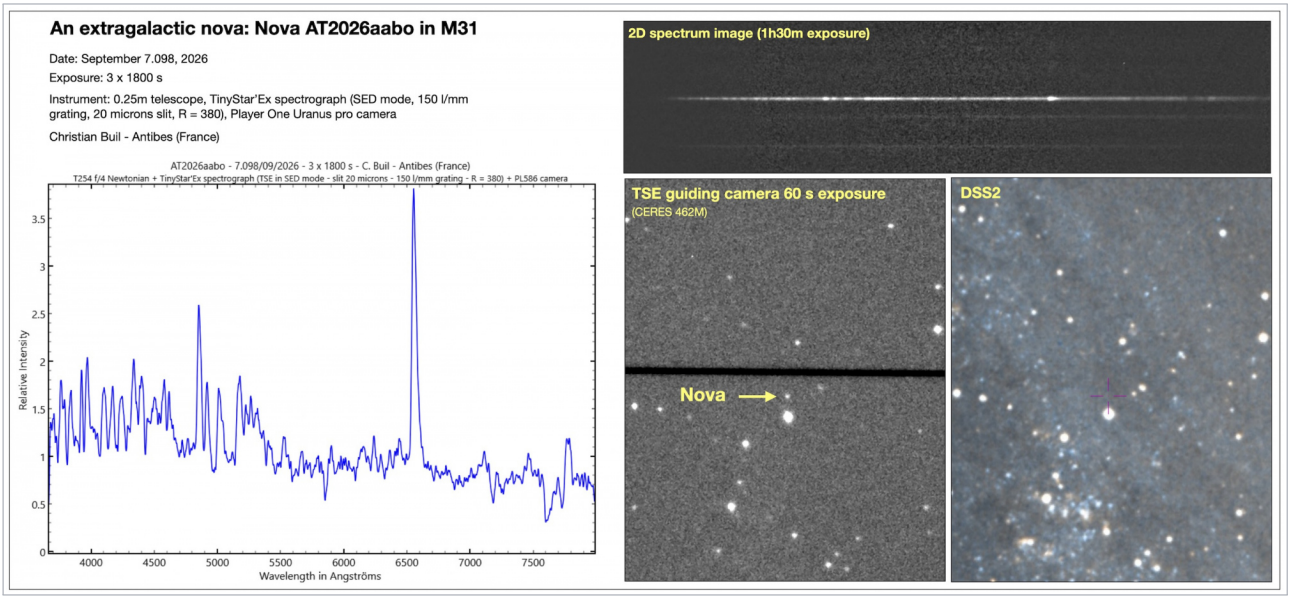
Spectrum of the symbiotic star Z And. TSE on a 150 mm f/5 Newtonian telescope. Guillaume Bertrand.



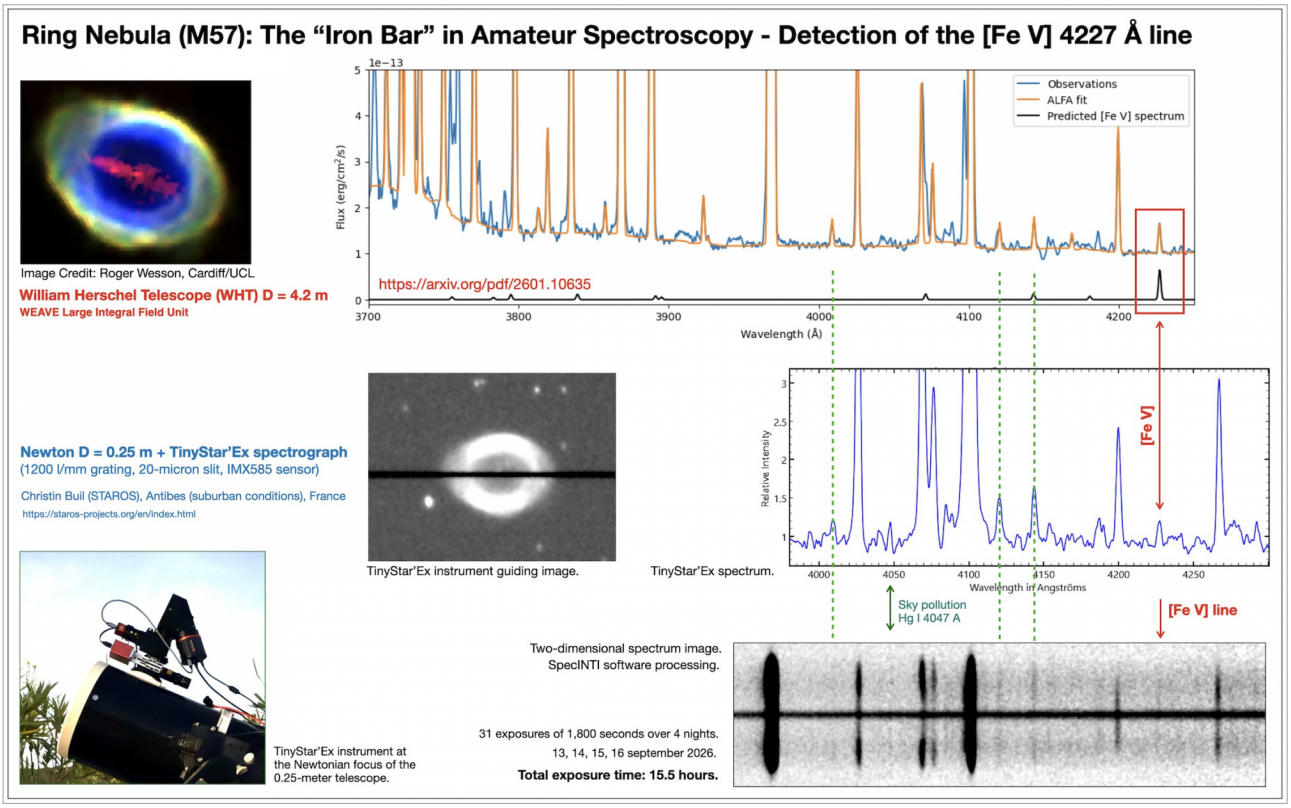
Spectrum of the nova V488 Sge. TSE on a 150 mm f/5 Newtonian telescope. Guillaume Bertrand.



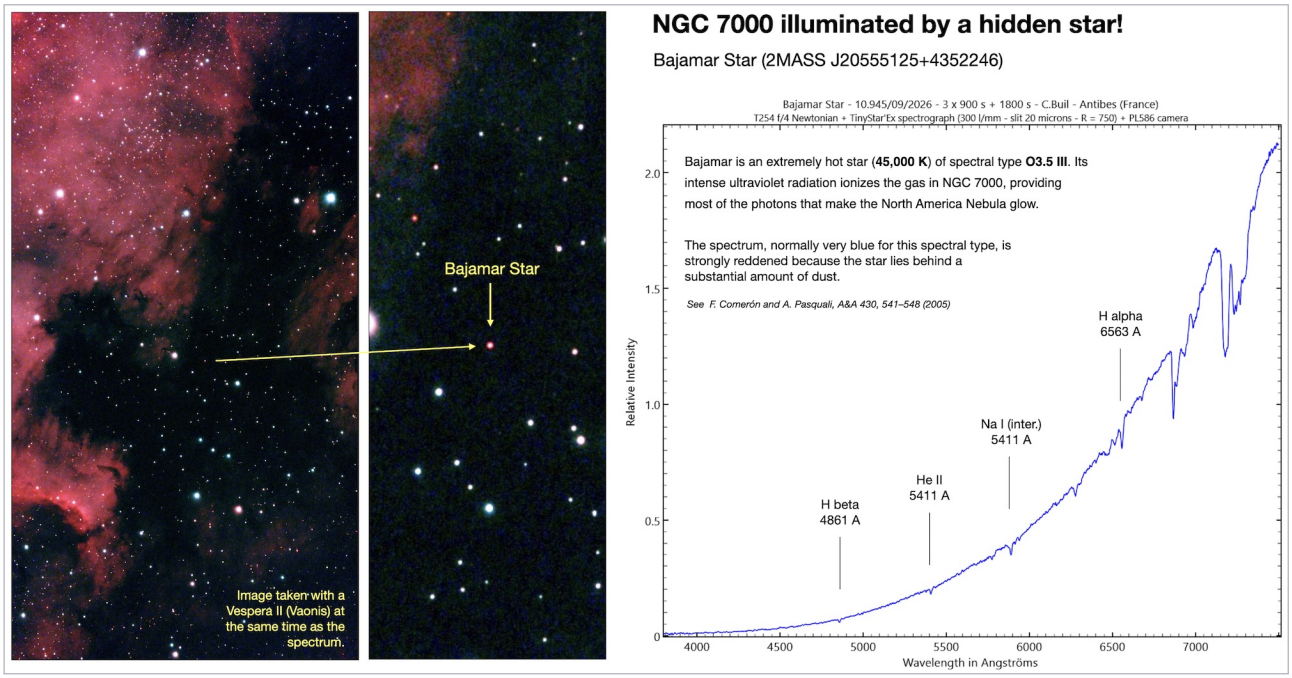
Spectrum of the extragalactic nova AT2026aabo (in Messier 31). TSE on a 250 mm f/4 Newtonian telescope. Christian Buil.



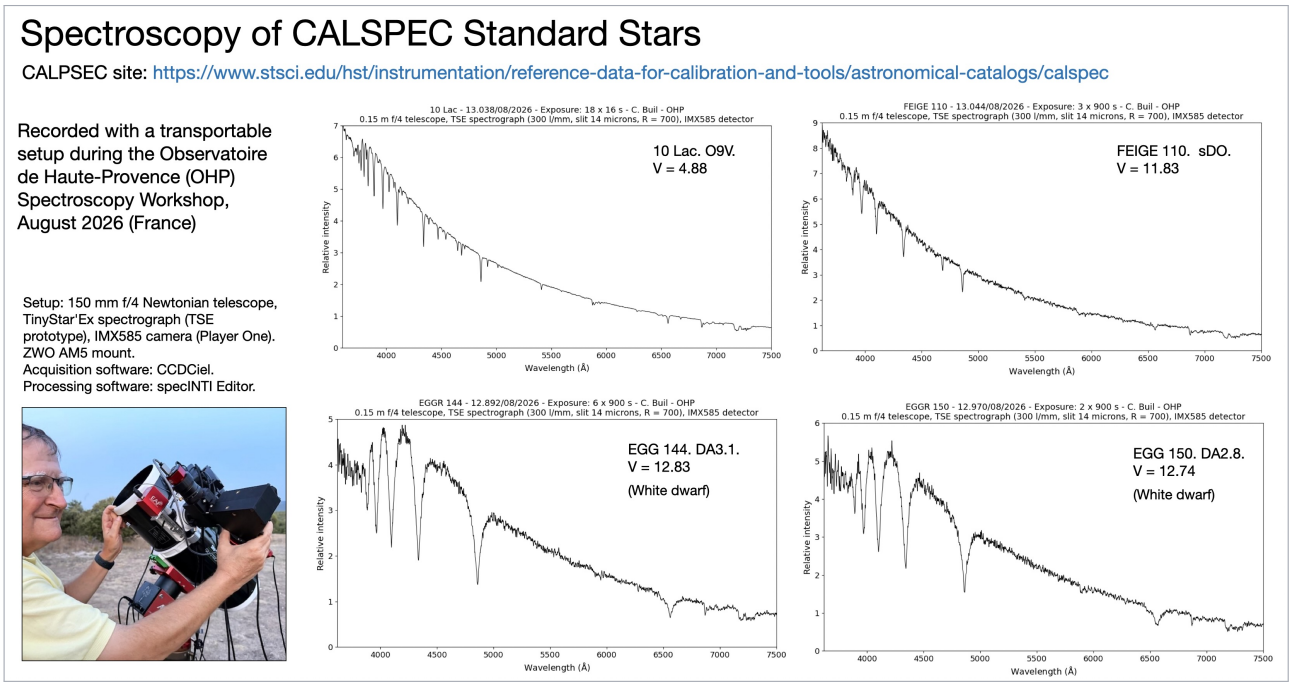
Spectrum of the recently discovered iron bar in Messier 57 (long exposure!). TSE on a 250 mm f/4 Newtonian telescope.



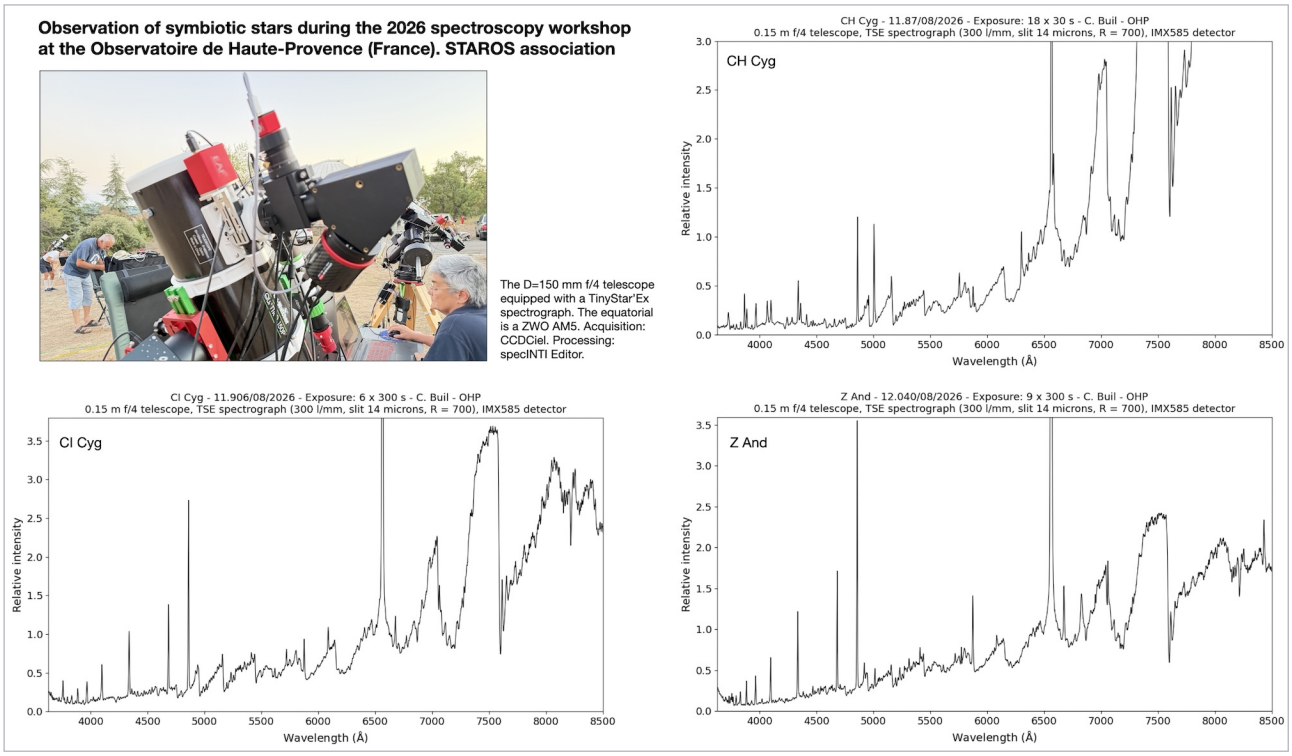
Spectrum of the hidden star that illuminates the large nebula NGC 7000 (North America Nebula). TSE on a 250 mm f/4 Newtonian telescope.



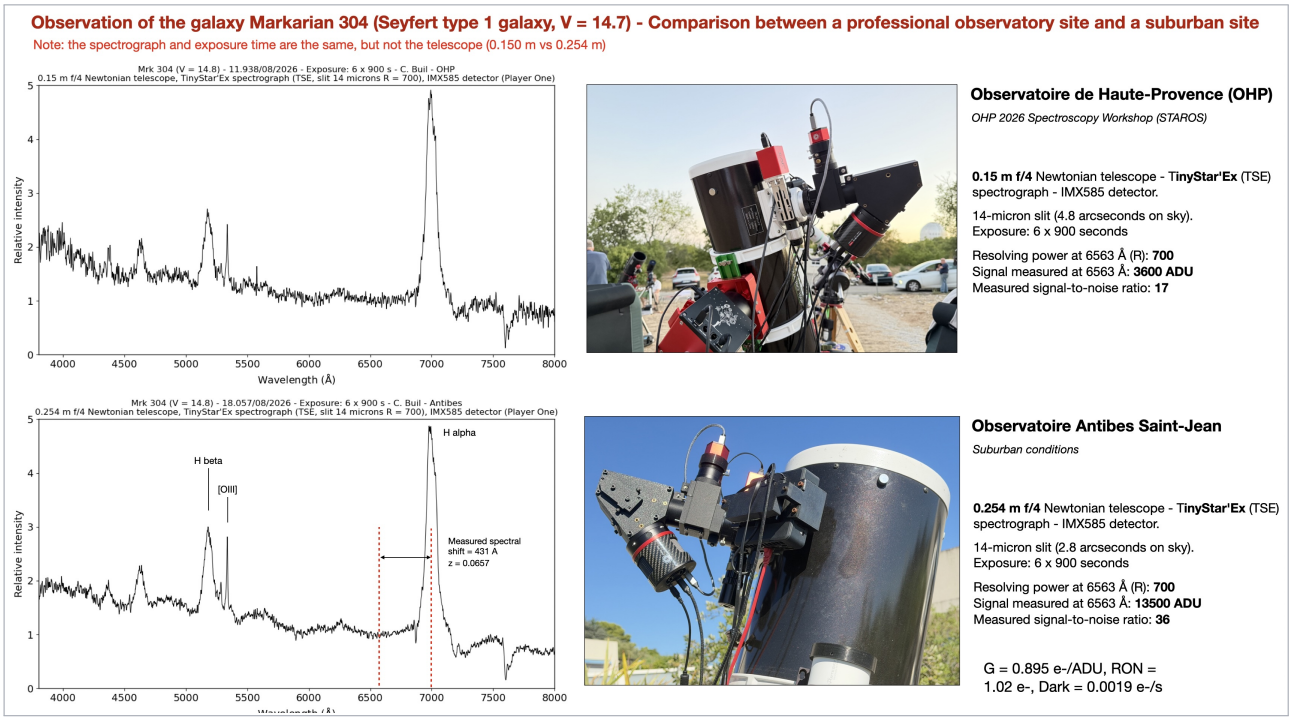
Spectra of spectrophotometric standards (very hot stars and white dwarfs). TSE on a 150 mm f/4 Newtonian telescope.



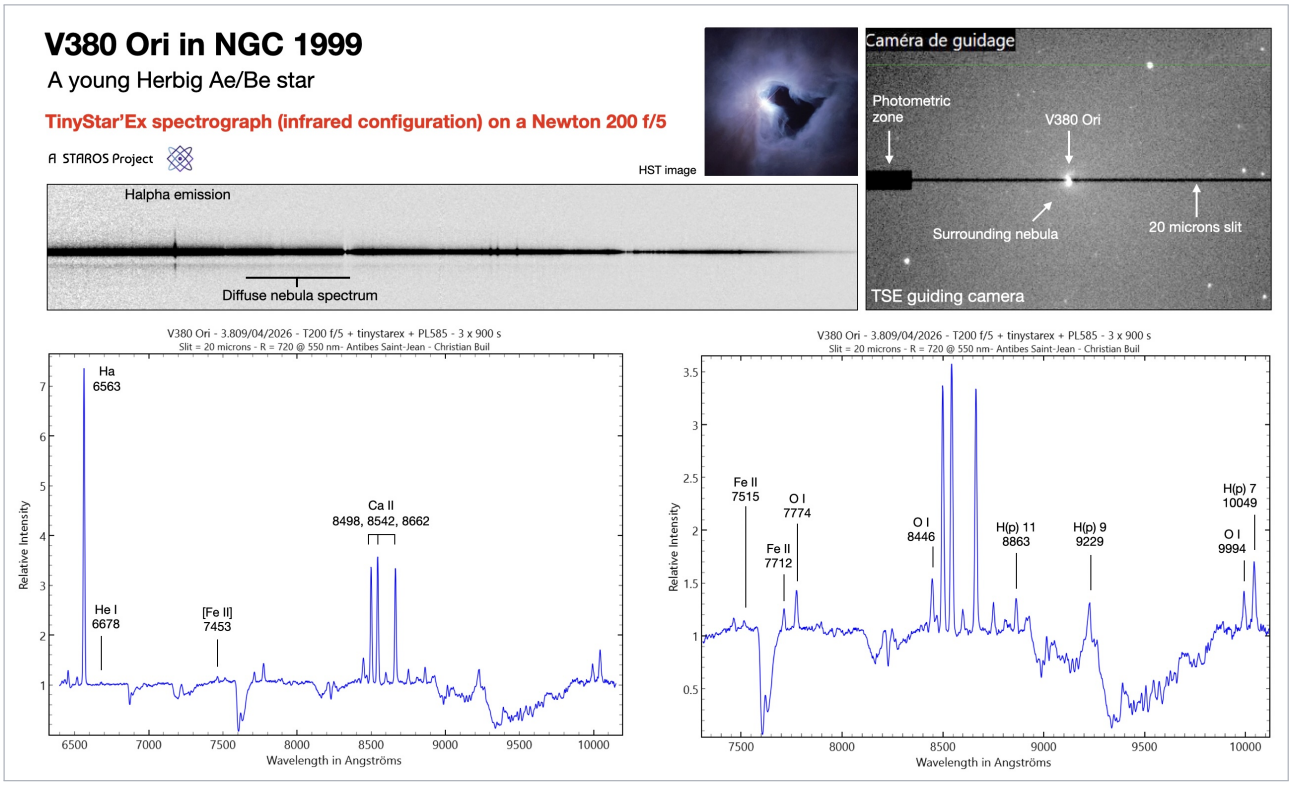
Spectra of symbiotic stars with a very compact telescope. TSE on a 150 mm f/4 Newtonian telescope.



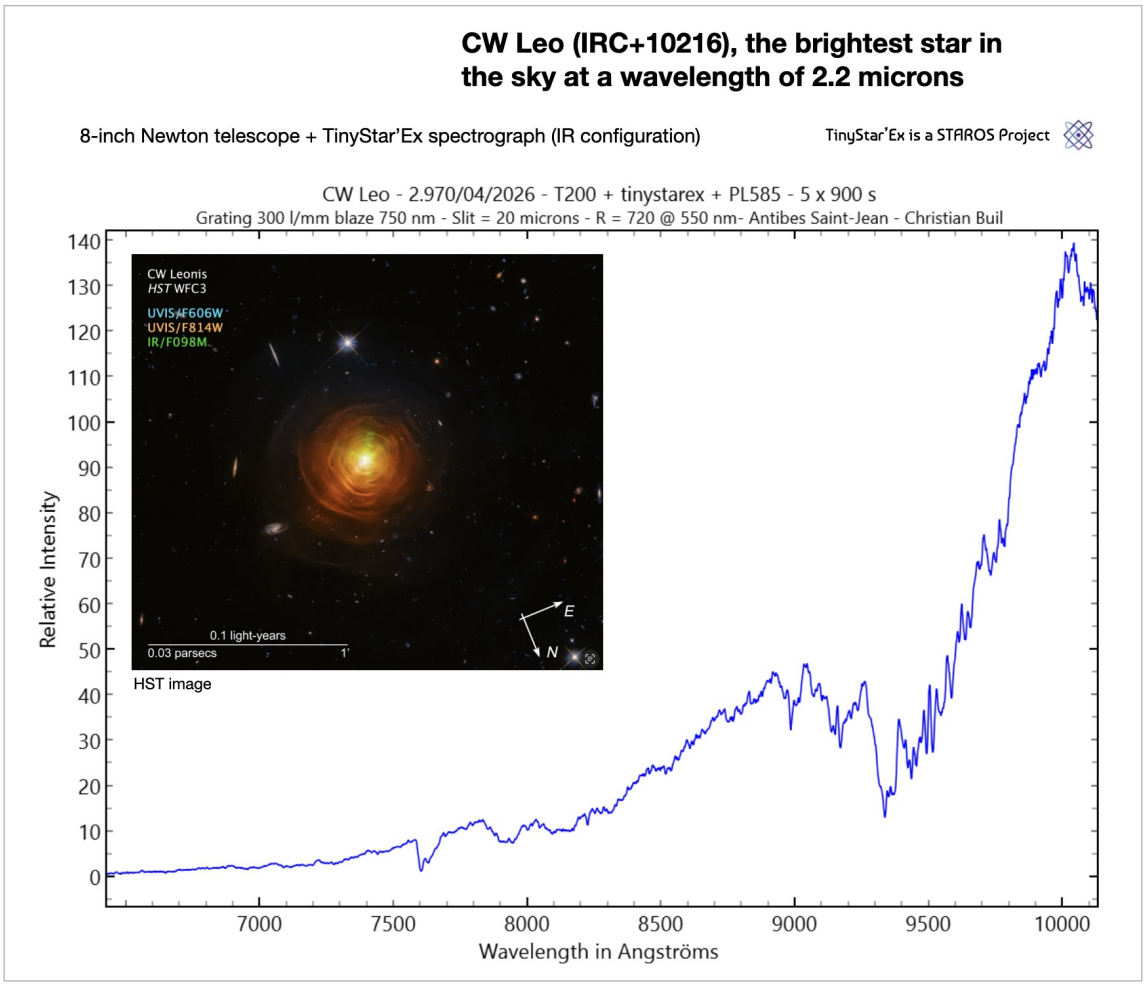
Spectrum of the galaxy Markarian 304 as a function of telescope aperture and observing site (with and without light pollution).
TinyStar'Ex spectra. Christian Buil.



Spectrum of the young star V380 Ori. It illustrates the excellent performance of TinyStar'Ex in the near infrared: the optics deliver a sharp image up to a wavelength of 1 micron, and even beyond.



CW Leo: the brightest star in the sky… in the infrared!



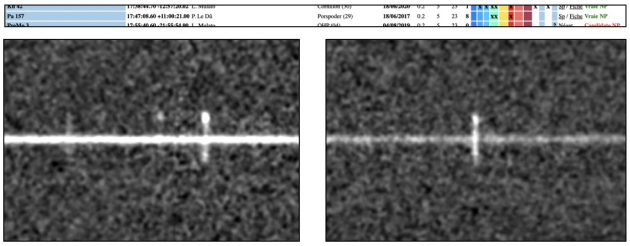
Observing the spectrum of an ultra-faint nebula from a city. Under these conditions, this object would remain invisible without a spectrograph.

TinyStar'Ex

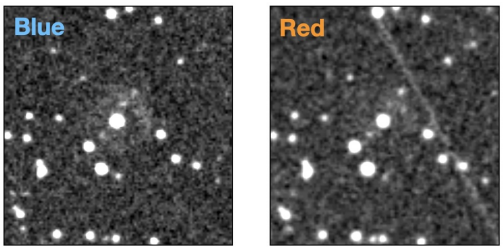
Pa 157: an ultra-faint planetary nebula

Amateur astronomers regularly discover new planetary nebulae.

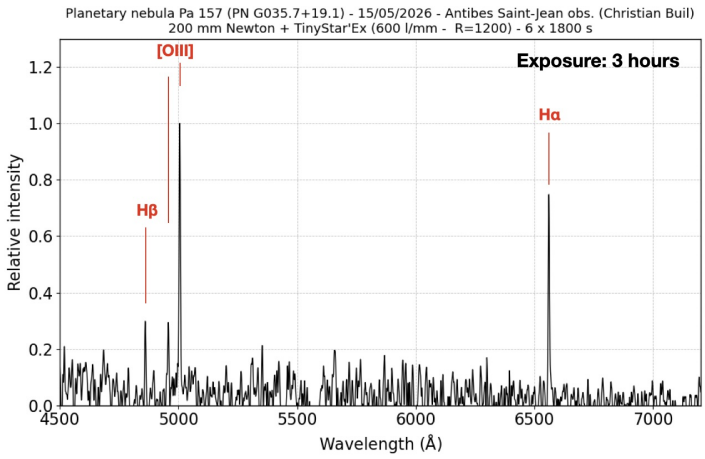
See **Planetary Nebulae.net** website: offers all the useful and practical information about the planetary nebulae (candidates), how to apply for a new discovery and how to confirm the candidates.

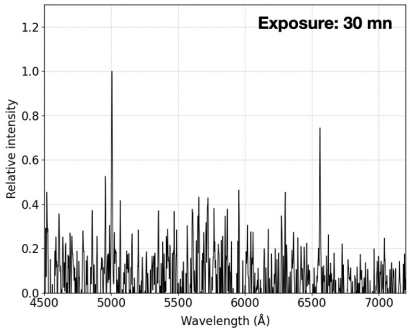


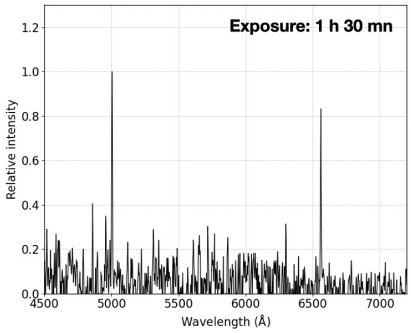
Extracts from the spectrum of the nebula Pa 157 (left: the [O III] and Hβ emission lines; right: the Hα line). The ring structure is clearly detectable, as well as the spectrum of the central star (V = 16.2). Total exposure time: 3 hours. Suburban observing conditions. 200 mm telescope + TinyStar'Ex (at R = 1200 configuration) + Player One Uranus M Pro.



DSS2 images of Pa 157: the nebula is barely detectable
RA: 17:47:09 - DEC: +11:00:21 (2000.0)







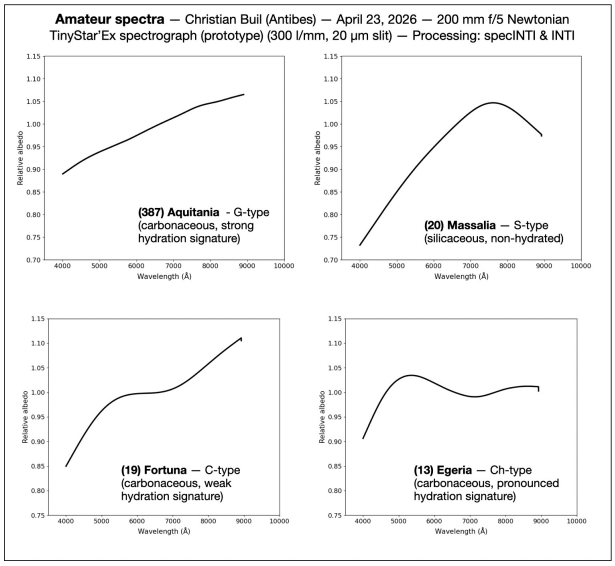
The spectra of asteroids.

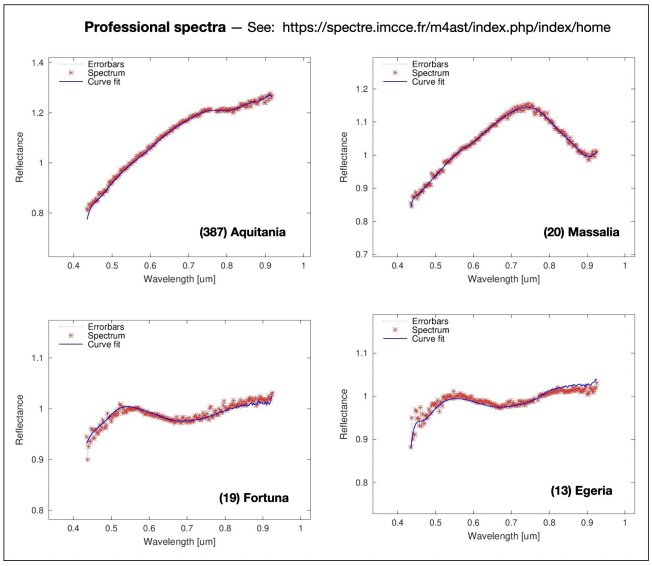
TinyStar'Ex project

Education and Science

Observing asteroids: study of spectral classes (or spectral taxonomy)

The diversity of observed spectral types largely depends on the chemical and mineralogical composition of the materials present on asteroid surfaces. This approach, which relates to the study of their geology, is accessible to amateur astronomers and offers a field of observation that is both original and scientifically rewarding.





Spectroscopy of everyday scenes. Using photographic or video lenses.



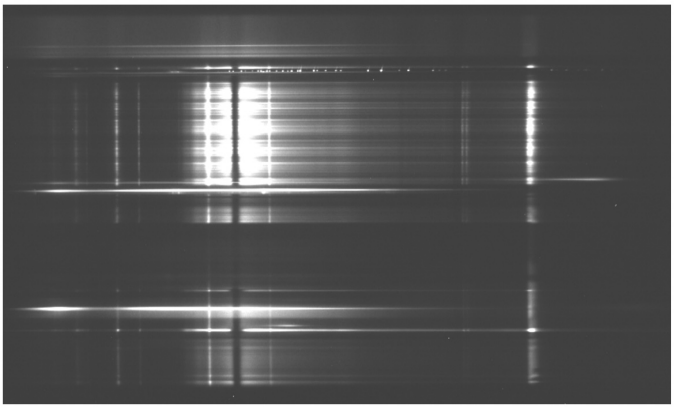
The spectral nature of the Eiffel Tower lighting.



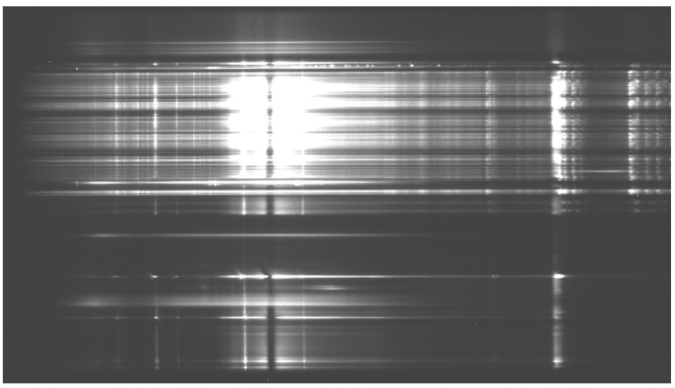
The Eiffel Tower and TinyStar'Ex

F=8.5 mm lens

25° Fied of View

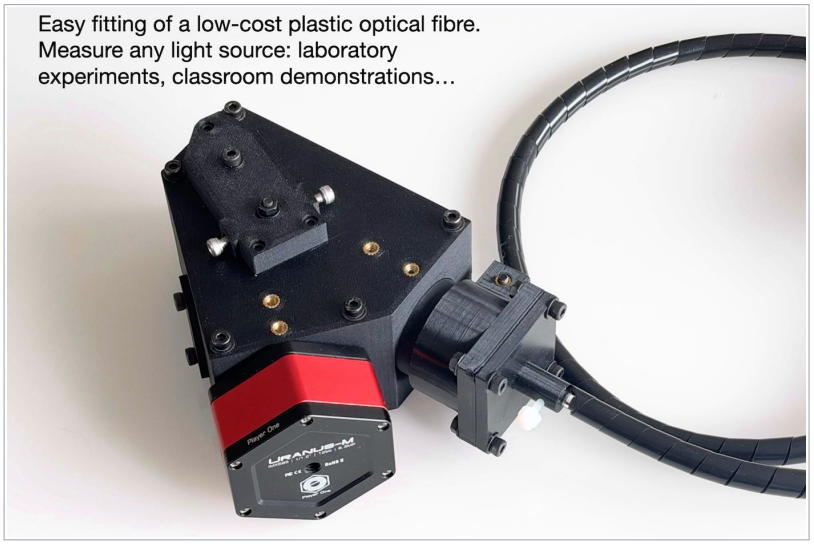


Spectrum of the Eiffel Tower's night lighting

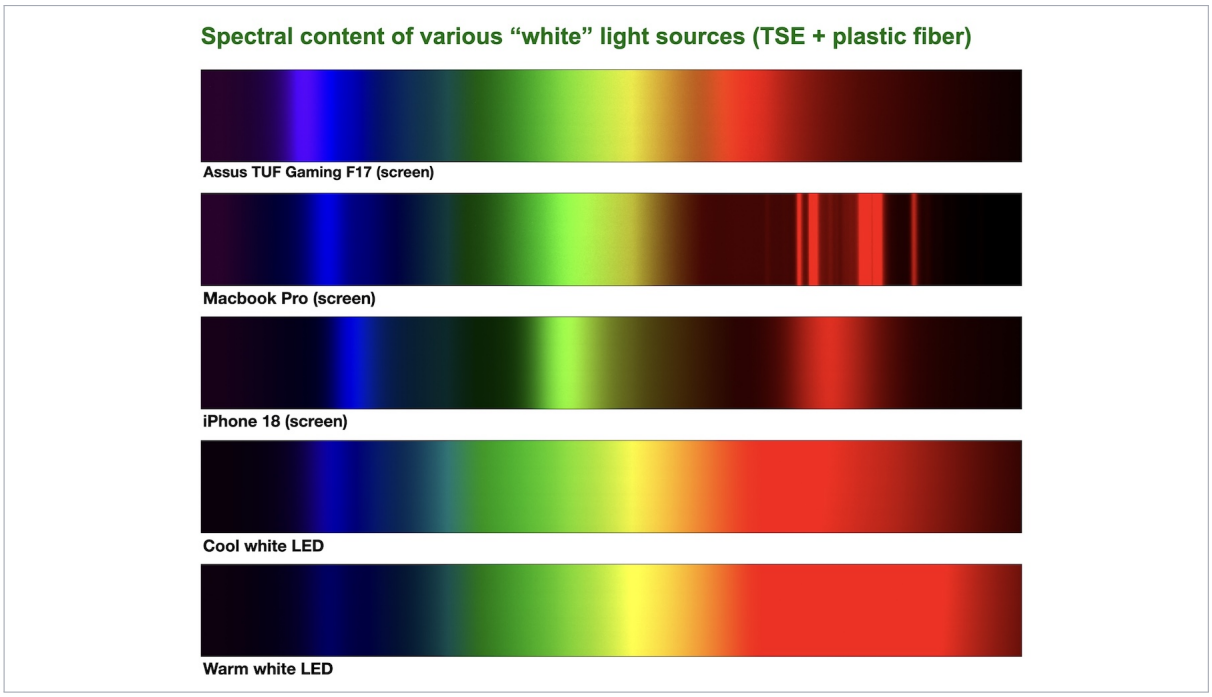


Spectrum of the Eiffel Tower's sparkling lights at the top of the hour

3D-printed adapter for illuminating the entrance slit with the output of an optical fibre. Here, a very inexpensive plastic light guide, 5 mm in diameter, sold for decorative lighting and easily cut to the required length with an ordinary pair of scissors. Ideal for physics experiments, at school or elsewhere, for collecting light from a wide variety of sources for spectral analysis, or simply for getting to grips with the techniques of spectroscopy.



Example of use of an optical fibre: comparison of the spectra produced by screens and LED lamps. The spectrum images were colourized afterwards. For this demonstration, the spectral response of the instrument has not been corrected: these are therefore raw data, as seen live, without any processing.



Observing very faint targets from a city, with an uncooled camera. Compact, affordable spectroscopy, anywhere!

TinyStar'Ex (TSE)

The spectrum of a magnitude 13.5 supernova (SN2026fvx in NGC 4205) with an uncooled CMOS camera

Cooled camera

Temperature: -20° C

Model: Player One Uranus-M pro
Sony IMX585 sensor

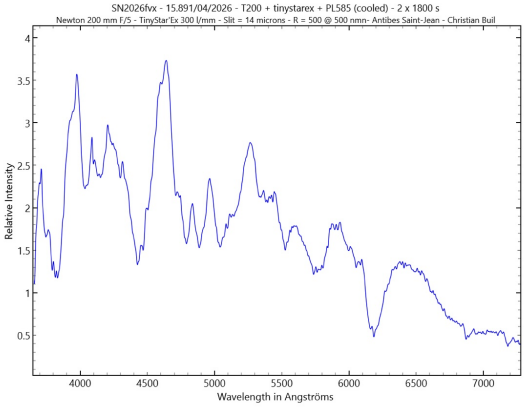
Gain = 220; Offset = 20; Effective gain = 0.911 e^-/ADU; Readout noise = 0.96 e^-;
Dark current = 0.00075 e^-/s

TinyStar'Ex spectrograph on 200 mm f/5 Newton telescope, slit GEN2 14 microns, 300 l/mm grating

Total exposure time: 2 x 1800 s (1 hour)
Signal to noise ratio: 150 @ 550 nm

April 15, 2026 - Antibes (France)





Uncooled camera

Temperature: +16° C

Model: Player One Uranus-M
Sony IMX585 sensor

Gain = 220; Offset = 20; Effective gain = 0.911 e^-/ADU; Readout noise = 1.03 e^-;
Dark current = 0.226 e^-/s

TinyStar'Ex spectrograph on 200 mm f/5 Newton telescope, slit GEN2 14 microns, 300 l/mm grating

Total exposure time: 20 x 180 s (1 hour)
Signal to noise ratio: 16 @ 550 nm

April 16, 2026 - Antibes (France)

